

Depth Matters but the Final Layer Suffices: Leakage-Free Evaluation of Layer Choice in Protein Language Models for Mutation Effect Prediction

Yutao Guo,^{1,2} Zihan Zhang,^{1,2} Xuezhou Zhao,^{1,2} Mengxi Chen^{1,2} and Dan Wu^{1,3,*}

¹International College of Pharmaceutical Innovation, Soochow University, Suzhou, 215123, China

²School of Pharmacy and Biomolecular Sciences, RCSI, Dublin, D02 VN51, Ireland

³Department of Chemistry, RCSI, Dublin, Dublin, D02 YN77, Ireland

*Corresponding author. danwu@suda.edu.cn

Abstract

Motivation: Protein language models such as ESM-2 encode biochemical information across 33 transformer layers. Multi-layer embedding fusion is often used under the assumption that different layers provide complementary mutation-relevant information. However, the conditions under which this assumption holds for mutation effect prediction remain poorly characterized. **Results:** We benchmarked two 650M-parameter protein language models (ESM-2 and the structure-aware SaProt) on the ProteinGym substitution benchmark using per-layer probing under three feature arms and four train–test splits, multi-scale pooling, logit-lens scoring, layer-wise similarity analysis, and a position-lookup control. Under the standard random split, all 33 layers gave nearly indistinguishable predictions ($\rho \approx 0.60$), but a control regressor given only random position vectors reproduced this level ($\rho = 0.600$) and collapsed to 0.10–0.14 under leakage-free splits (paired drop -0.46 , Wilcoxon $p = 5.2 \times 10^{-12}$): random-split layer interchangeability is largely position memorization. Under the three leakage-free splits (modulo, contiguous, position-grouped), performance rose with depth for both models (mutation-specific embeddings up to $\rho = 0.53$), the best layer lay in a narrow late band (layers 30–32), and the best layer exceeded the final layer by at most $\Delta\rho = 0.009$; across the 20 non-degenerate best-versus-final comparisons, 8 were equivalent at the ± 0.01 margin (bootstrap 90% confidence interval inside the margin), 12 were inconclusive, and none exceeded the margin; in the remaining 7 configurations the best layer was the final layer itself. A single-substitution-only sensitivity analysis reproduces the depth trend and the structure-token benefit, with the final layer remaining within 0.01 ρ of the best layer for SaProt and within 0.02 for ESM-2. A logit-lens analysis validated against the models' official logits pathway showed the same depth trend zero-shot (final layer best: ESM-2 $\rho = 0.443$, SaProt $\rho = 0.437$), consistent with published ProteinGym zero-shot levels. Real Foldseek 3Di structure tokens improved SaProt under every split—slightly under random ($+0.003$ to $+0.032$) and substantially under leakage-free evaluation ($+0.06$ to $+0.12$, all $p \leq 10^{-4}$). No fusion strategy improved on the final layer under leakage-free evaluation (paired differences $\leq +0.018$ ρ , none significant; several significantly negative with nonlinear readouts), whereas the same comparisons under the random split showed small significant advantages for fusion ($+0.01$ to $+0.05$): the apparent benefit of multi-scale pooling is protocol-dependent and does not survive leakage-free evaluation. Switching the readout mattered more than fusing layers, but only for ESM-2 (last-layer LightGBM $+0.045$ over Ridge, $p = 0.011$). Depth is important for mutation effect prediction, the final layer is already sufficient, and random-split evaluation masks both facts. **Availability and Implementation:** All code and per-dataset results are available at <https://github.com/schwarzeteeguo-creator/esm2-layer-invariance>. An archived snapshot of the code and per-dataset results for this revision is deposited at Zenodo (DOI: 10.5281/zenodo.22844290). The benchmarking pipeline, including masked-position feature extraction, multi-scale pooling, and logit lens analysis, is implemented in Python using PyTorch and scikit-learn.

Keywords protein language models, layer selection, final-layer sufficiency, mutation effect prediction, ProteinGym, position leakage, logit lens, CKA

1. Introduction

Protein language models (PLMs) pre-trained via masked language modeling (MLM) on millions of protein sequences have become indispensable for mutation effect prediction , . Among these, ESM-2 is one of the most widely adopted, with a

650M-parameter architecture comprising 33 transformer layers that produce 1280-dimensional per-residue hidden states. These embeddings have been successfully applied to diverse tasks including variant effect prediction , , protein design, and functional

annotation , , establishing ESM-2 as a foundational tool in computational biology.

The internal organization of information across transformer layers has been extensively studied in natural language processing, where the “BERTology” literature , has revealed that different layers specialize in distinct linguistic phenomena: lower layers encode surface-level syntax, middle layers capture semantic content, and final layers refine task-specific predictions . This hierarchical view has motivated analogous investigations in protein language models. A growing body of work posits that different transformer layers in PLMs similarly encode complementary biochemical information, and that combining representations from multiple layers should therefore improve downstream tasks. Kumar and Jha demonstrated that mid-to-late ESM-2 layers (20–33) outperform the final layer for kinase function prediction, while early layers (6–14) have been shown to encode local physicochemical properties such as hydrophobicity and secondary structure propensity . Alsamkary, Elshaffei et al. proposed hierarchical pooling and pooled attention architectures for fusing representations across multiple PLMs and reported up to a 12% Spearman improvement for protein–protein interaction prediction. More broadly, the strategy of extracting and fusing embeddings from multiple transformer layers or multiple PLMs has become common practice in PLM-based pipelines , , often under the implicit assumption that each source contributes non-redundant information.

These claims, however, rest on an assumption that has not been systematically tested: that different ESM-2 layers genuinely encode *complementary* information for mutation effect prediction, rather than merely offering different views of the same underlying signal. If layers are functionally similar under a given readout and evaluation protocol, then multi-scale fusion may add cost without improving prediction. If layer complementarity appears only under specific readouts or split protocols, then layer selection should be treated as a conditional design choice rather than a universal rule. Resolving this question carries practical consequences: the computational cost of extracting and storing embeddings from multiple layers is substantial for large-scale screening efforts, and the proliferation of ad-hoc layer-selection heuristics complicates reproducibility.

To resolve this question, we conducted a large-scale benchmark on ProteinGym , a comprehensive collection of deep mutational scanning (DMS) experiments spanning diverse protein families, assay types, and organismal contexts. We employed three independent experimental paradigms (Figure 1): (1) per-layer supervised probing via Ridge regression, (2) multi-scale embedding pooling with controlled feature sets, and (3) logit lens zero-shot scoring. In parallel, we quantified layer-wise representational similarity through cosine and CKA analysis. Together, these approaches ask how much mutation-relevant information each transformer layer carries, how that information grows with depth, and how the answer depends on the evaluation protocol. Our findings revise the premise of the multi-scale-fusion literature on three points. First, the flat per-layer profiles produced by the standard random split are largely position memorization: a lookup-table control reproduces them entirely. Second, under leakage-free evaluation, predictive information rises with depth, but the final layers are interchangeable among themselves—the last layer is within $\Delta p \leq 0.01$ of the best

layer everywhere we measured—so depth matters, yet extracting the final layer alone is sufficient. Third, no multi-scale fusion strategy improves on the final layer under either evaluation protocol, whereas structure tokens contribute a genuine, measurable benefit that only leakage-free evaluation can see.

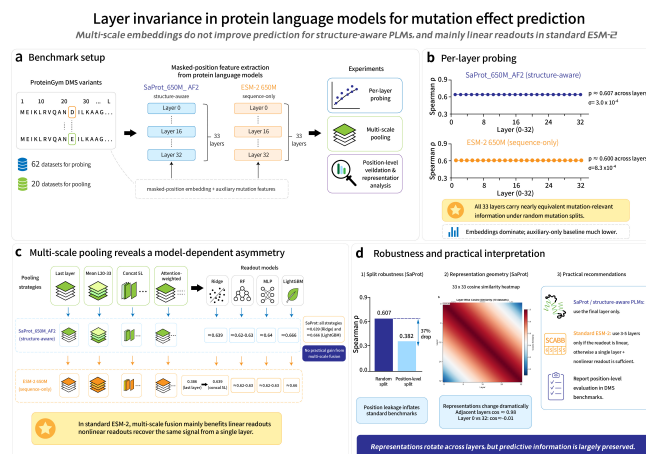


Figure 1 Experimental design overview. The three complementary experimental paradigms employed in this study: (a) Per-layer probing: Ridge regression on hidden states extracted from each of the 33 transformer layers under three feature arms (masked wild-type, unmasked wild-type, and mutant) and four train–test splits (random, modulo, contiguous, and position-grouped), quantifying the predictive information content per layer together with a random-position lookup control. (b) Multi-scale pooling: four embedding fusion strategies (Last Layer, Mean L20–33, Concat 5L, Attention-weighted, with attention weights trained within each cross-validation fold) tested across four readout models (Ridge, Random Forest, MLP, LightGBM) to assess whether combining layers improves prediction. (c) Logit lens: applying the final-layer LM head to intermediate hidden states for zero-shot mutation scoring, directly measuring how token predictions are refined across depth. (d) Layer-wise cosine similarity analysis: quantifying representational divergence independent of predictive performance. Together, these paradigms provide converging evidence on whether different transformer layers encode complementary or redundant information for mutation effect prediction.

2. Methods

2.1. Models and Data

We used SaProt_650M_AF2 , a structure-aware variant of ESM-2 650M that extends the vocabulary to 446 structure-informed tokens (21 amino acids $\times$ 21 Foldseek 3Di states), as our primary model. For validation, we also tested standard ESM-2 650M in both the probing and logit lens experiments, enabling direct comparison between structure-aware and sequence-only representations. All experiments used ProteinGym’s substitution DMS benchmark . The reference release is ProteinGym v1.3, which contains 217 substitution assays; our experiments use the 63-assay LMDb snapshot that the SaProt repository distributes for supervised fitness prediction (<https://github.com/westlake-repl/SaProt>). All 63 assays are contained in the v1.3 benchmark, the subset was fixed before any experiment

and is identical across all arms, models, and splits, and the full assay list is tabulated in Supplementary Table S1. Because each assay contributes one train–test problem, all n refer to assays: $n = 63$ for the random, modulo, and position-grouped splits; six datasets whose mutated positions form a single contiguous block have undefined Spearman correlations under the contiguous split, which is therefore reported on the remaining 57. The pooling benchmark was run on a computationally tractable subset of 20 datasets because it crossed four pooling strategies with four readout models and multiple cross-validation folds. We therefore treat the pooling analysis as a controlled validation of the probing result rather than as a full ProteinGym-wide estimate. The subset comprises the first 20 datasets in alphabetical order, fixed identically for both models. Per-dataset mutation counts range from under 100 to over 200,000. Each DMS variant is exploded into one row per substituted position (multi-site variants are common in this benchmark: they are 84.6% of variants and contribute 95.8% of all analyzed rows; Supplementary Table S1). Datasets were capped at 10,000 variants before row construction (11 of 63 datasets) and at 5,000 rows inside the pipeline (26 of 63 datasets reach this ceiling), both with a fixed seed shared across arms, splits, and layers. For SaProt, we additionally prepared real Foldseek 3Di structure tokens by assigning each wild-type residue the 3Di state of its aligned residue in the AlphaFold Protein Structure Database entry for that sequence (structures available for 62 of 63 datasets; two aligned structures were truncated at the C terminus and the missing residues were padded with the Foldseek gap state; the influenza nucleoprotein dataset has no AFDB entry and was evaluated with structure-masked tokens). SaProt results are reported for both structure-masked and real-3Di inputs (per-dataset 3Di file lengths and gap counts in Supplementary Table S5).

2.2. Experiment 1: Per-Layer Probing

For each DMS dataset we extracted per-layer hidden states under three feature arms. The **masked.wt** arm masks each mutated position in its own forward pass of the wild-type sequence (“M#” for SaProt, <mask> for ESM-2; single-position masking), and each row receives the hidden state at that row’s substitution position—the conventional masked-position probing protocol. The **unmasked.wt** arm reads the wild-type token at the mutated position without masking (one forward pass per dataset, gathered at all mutated positions). The **mutant** arm feeds the full mutant sequence and reads the hidden state at the mutated position, averaged over the variant’s mutated positions for multi-site variants; it is the only arm whose representation differs between two mutations at the same position, and therefore the only one that can carry mutation-specific information beyond position identity. Both SaProt.650M_AF2 (structure-masked and real-3Di inputs) and standard ESM-2 650M were evaluated under identical conditions.

Each arm was evaluated under four train–test splits, all with five cross-validation folds (four for the single four-position dataset under the position-grouped split). The **random** split assigns whole variants to folds at random (seeded), so all rows of one variant always share a fold. The **modulo** split assigns each residue position to fold $(\text{position} - 1) \bmod 5$ and the **contiguous** split divides the sequence into five contiguous segments, both following the official ProteinGym protocol; under these

schemes a row’s fold is determined by its own substitution position, so the rows of a multi-site variant can fall into different folds (1.10M of 1.30M multi-site variants straddle modulo folds; 0.36M straddle contiguous folds). These splits are leakage-free with respect to positions but retain variant-level label sharing for multi-site variants; a single-substitution-only sensitivity analysis (Section 3.8) therefore re-runs the mutant arm restricted to single-site variants. Finally, a **position-grouped** split (GroupK-Fold over residue positions, so that all mutations at a given position share a fold) serves as a strict control. For each layer $l \in \{0, \dots, 32\}$ and feature configuration (combined: the hidden state concatenated with 42 auxiliary features, namely one-hot encodings of wild-type and mutant amino acids, the BLOSUM62 substitution score, and a normalized Grantham distance proxy; embedding_only; and auxiliary_only), we trained Ridge regression with the regularization strength selected per dataset by generalized cross-validation (grid 10^{-2} to 10^3), standardization fit inside each training fold, and reported Spearman rank correlation ρ . The single dataset with fewer than five mutated positions (four positions) used four position-grouped folds.

To quantify how much probing performance is attributable to position identity alone, we added a **random-position control**: each mutation is represented by a fixed random vector indexed by its residue position (33 independent draws), probed with the identical Ridge protocol with and without auxiliary features. This control contains no sequence information; any performance it recovers measures position memorization.

2.3. Experiment 2: Multi-Scale Pooling

We compared four embedding strategies using fixed auxiliary features, evaluated on the **mutant** arm under the modulo split as the primary leakage-free protocol and under the random split as a protocol contrast: (1) **Last Layer**: the standard approach using only the final transformer layer; (2) **Mean L20–33**: mean pooling across layers 20–33, motivated by Kumar and Jha ; (3) **Concat 5L**: concatenation of layers 6, 14, 20, 26, and 32 (0-indexed, including the final layer), spanning the full representational depth; and (4) **Attention-Weighted**: softmax-weighted pooling across all 33 layers, with the weights trained within each cross-validation fold (120 Adam steps jointly with a closed-form ridge readout on training rows only, seeded).

For each strategy, we evaluated Ridge regression (regularized by generalized cross-validation as above), Random Forest (50 estimators, max depth 10), MLP (two hidden layers of 256 units, ReLU), and LightGBM (100 estimators, max depth 7). Every dataset and strategy used the same five outer cross-validation folds as the probing experiment; standardization was fit inside training folds, and the auxiliary features were identical across strategies, isolating the effect of embedding choice.

2.4. Experiment 3: Logit Lens

We applied the model’s final-layer LM head to the hidden states of each intermediate layer, producing pseudo-logits for zero-shot mutation scoring. All computations ran in fp32, and for every dataset the final-layer pseudo-logits were validated against the model’s official logits pathway (maximum $|\Delta \log P| < 10^{-3}$; all 63 datasets passed for both models); no per-layer trend is interpreted except through gate-passed results (cf. the tuned

lens literature). For standard ESM-2, we used the conventional log-ratio, $\log P(\text{mutant}) - \log P(\text{wild-type})$. For SaProt with structure-masked input tokens (AA#, where # denotes masked structure), we marginalized over all 21 structure-token variants per amino acid; with real 3Di inputs, the wild-type structure token at each position was retained. We reported per-layer Spearman ρ between these zero-shot scores and experimental DMS fitness values across all 63 datasets.

2.5. Layer Similarity Analysis

We computed the 33×33 cosine similarity matrix of per-layer hidden states, averaged over 200 randomly selected residue positions across 10 datasets (a 50-position variant is retained for continuity with the original analysis; the larger sample reduces the known small-sample upward bias of CKA). Because raw cosine is not invariant to rotations of the coordinate system, we additionally computed linear CKA and the debiased (unbiased-HSIC) linear CKA estimator on the same hidden-state matrices. The unit of analysis for similarity-versus-distance statistics is the set of unique layer pairs (i, j) with $i < j$, diagonal excluded. A high CKA combined with a near-zero cosine indicates a change of basis that preserves linearly decodable information.

2.6. Statistical Evaluation

We used three complementary procedures: (1) an **invariance score**, defined as the standard deviation of per-layer mean Spearman ρ (smaller indicates a flatter layer profile); (2) **equivalence testing**: for each model, arm, and leakage-free split, we tested whether the mean per-dataset difference between the best layer and the final layer falls within a margin of $\Delta\rho = 0.01$. The margin was fixed before the equivalence analysis as the smallest difference that could plausibly change a downstream layer-choice decision; it is a reporting convention, not a task-derived value, and is applied identically everywhere. The best layer was selected as the argmax of the layer-mean curve computed on the same per-dataset results; this same-data selection biases the procedure towards detecting a difference, which makes any equivalence verdict conservative. A comparison was declared *Equivalent* when the bootstrap 90% confidence interval of the paired mean difference (10,000 resamples, seeded percentile method) lay entirely within ± 0.01 , a *Difference* when it lay entirely outside, and *Inconclusive* otherwise; two one-sided tests (TOST) p -values and Cohen's d_z are reported alongside. In 7 of the 27 model-arm-split combinations the best layer was the final layer itself (no test defined); the remaining 20 comparisons constitute Supplementary Table S4; and (3) **paired Wilcoxon signed-rank tests** for split contrasts (random versus leakage-free), for real-3Di versus structure-masked SaProt inputs, for each fusion strategy versus the last layer, and for readout contrasts, each reported with bootstrap 90% confidence intervals of the paired mean difference.

3. Results

3.1. Representations Rotate While Linearly Decodable Information Is Preserved

We first asked whether different transformer layers encode representations that are merely different views of the same underlying information, or whether they capture genuinely complementary features. To answer this, we computed the 33×33 layer-wise cosine similarity matrix of per-layer hidden states, averaged over 50 randomly selected residue positions across 10 datasets (Figure 2).

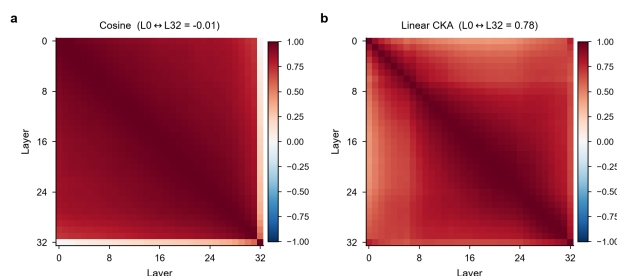


Figure 2 Layer-wise similarity matrices for SaProt (33×33). Averaged over 10 datasets and 200 random residue positions per dataset. (a) Cosine: adjacent layers are highly similar ($\cos = 0.98$); mean similarity decays monotonically with distance and collapses to -0.01 only between the most distant layers (Layer 0 vs. 32). (b) Linear CKA over the same hidden states: the most distant layers retain CKA 0.81, indicating a rotation that preserves linearly decodable information rather than information loss.

The matrix revealed a smooth representational trajectory across layers. Adjacent layers were highly similar (mean cosine 0.96–0.98), and cosine decayed monotonically with layer distance, gently at first and steeply only at the largest distances (Pearson r on unique layer pairs: -0.93 for ESM-2, -0.50 for SaProt). For SaProt the mean cosine by distance fell from 0.97 ($d = 2$) through 0.90 ($d = 10$) and 0.81 ($d = 20$) to 0.50 ($d = 30$), collapsing to -0.008 between the most distant layers (Layer 0 vs. Layer 32); ESM-2 decayed more gently and remained positive throughout (0.93 at $d = 2$, 0.58 at $d = 31$, 0.54 at $d = 32$). Crucially, near-zero raw cosine does not imply lost information: the linear CKA between SaProt layers 0 and 32 was 0.78 (0.81 in the 50-position variant) while their cosine was -0.008 . The CKA-distance profile is non-monotonic for SaProt: CKA dips to 0.67 around $d = 20$ and rises again to 0.78 at $d = 32$, so the most distant layer pair retains more linearly shared information than mid-depth pairs—consistent with residual streams accumulating rather than discarding early-layer content. For ESM-2 the same pair gives CKA 0.41 with cosine 0.54 (adjacent-layer CKA 0.95–0.97 for both models). Increasing the sample from 50 to 200 positions lowered the ESM-2 distant-layer CKA from 0.55 to 0.41, confirming that small-sample CKA is upward-biased at this dimensionality; the debiased unbiased-HSIC estimator is unstable here ($n < d$; point estimates can exceed 1), so we rely on the larger-sample standard estimator

(both estimators at both sample sizes are tabulated in Supplementary Table S9). The depth-wise transformation is therefore well described as a rotation of the representation that preserves much linearly decodable information (Figure 3c). This large rotation of the representational geometry raises a sharp question: *does this representational transformation reflect complementary information encoded at different depths, or is mutation-relevant predictive performance preserved under this transformation?*

To answer this, we conducted three converging experimental paradigms, each designed to interrogate the information content of individual layers from a different angle.

3.2. Random-Split Layer Interchangeability Is Largely Position Memorization

We first asked whether the near-flat random-split layer profile reflects genuine representational redundancy. Under our pipeline the profile is exactly this flat: for the masked_wt arm, per-layer mean ρ spans 0.603–0.606 for both models with invariance scores $\sigma \approx 5 \times 10^{-4}$ to 1.2×10^{-3} (Table 2). A Ridge regressor given *random vectors indexed only by mutation position* plus the same auxiliary features recovered $\rho = 0.600$ under the random split (Table 1), matching the full masked-position PLM probing level. Under leakage-free splits the same control collapsed to $\rho = 0.138$ (modulo), 0.096 (contiguous, $n = 57$), and 0.135 (position-grouped), a paired drop of -0.46 (random versus modulo, Wilcoxon $p = 5.2 \times 10^{-12}$); the random vectors alone carried 0.557 under the random split and approximately zero elsewhere, and the auxiliary features alone fell from 0.339 to 0.19–0.23. Because every mutation at the same position shares one masked-position embedding, the random-split task is solvable by a position lookup table; the flat layer profile is therefore largely an artifact of position memorization rather than evidence that mutation information is distributed uniformly across depth.

Table 1. Random-position lookup control (63 datasets; contiguous $n = 57$). Mean Spearman ρ of Ridge regressors given no sequence information. The control reproduces the random-split PLM probing level and collapses under leakage-free splits.

Features	Random	Modulo	Contiguous
Random position vector + aux	0.600	0.138	0.096
Random position vector only	0.557	−0.008	−0.056
Auxiliary only	0.339	0.230	0.188

Table 2 and Figure 3a report the per-layer probing of all three arms, both models, and all four splits ($n = 63$; contiguous $n = 57$). Two patterns are stable across every leakage-free column. First, predictive information rises with depth: every leakage-free column improves monotonically from layer 0 to the late 20s, and the best layer sits in a narrow late band—layers 30–32 are best in 26 of 27 model–arm–split combinations (layer 27 in the remaining one), while in the other 7 combinations the best layer is the final layer itself. Second, the best layer is at most marginally better than the final layer: across the 20 non-degenerate best-versus-final comparisons the mean per-dataset difference never exceeded $\Delta\rho = 0.009$; with the unified verdict

rule (bootstrap 90% confidence interval of the paired mean difference entirely inside ± 0.01), 8 comparisons were equivalent, 12 were inconclusive, and none indicated a difference beyond the margin (Supplementary Table S4). Because the best layer is selected on the same data, this procedure is biased towards detecting a difference, making the equivalence verdicts conservative. The invariance score rose from $\sigma \approx 10^{-3}$ (masked_wt, random) to $4\text{--}7 \times 10^{-2}$ under leakage-free evaluation, yet remained well below the cross-dataset standard deviation (≈ 0.17): performance varies far more *across* datasets than *across* layers, but under leakage-free evaluation the residual layer variation has a consistent direction, favoring the final layers. The mutant arm exceeded the masked arms in every split (e.g., 0.681 versus 0.606 under random, 0.483 versus 0.459 under modulo for ESM-2), confirming that mutation-specific signal exists and is recoverable once the representation actually depends on the mutation.

Table 2. Unified per-layer probing, combined features (mean Spearman ρ ; best layer in parentheses, followed by the final-layer value when the best layer is not layer 32). $n = 63$; contiguous $n = 57$. Full 33-layer curves in Supplementary Table S2; per-dataset results in Supplementary Table S3.

Model	Arm	Random	Modulo	Contiguous	Pos.-grouped
ESM-2	masked_wt	0.606 (32)	0.459 (32)	0.370 (32)	0.450 (32)
ESM-2	unmasked_wt	0.606 (32)	0.470 (32)	0.388 (32)	0.461 (32)
ESM-2	mutant	0.681 (32)	0.483 (31) → 0.479	0.365 (31) → 0.360	0.477 (31) → 0.475
SaProt (mask)	masked_wt	0.603 (31) → 0.603	0.391 (30) → 0.386	0.282 (31) → 0.274	0.385 (31) → 0.382
SaProt (mask)	unmasked_wt	0.604 (32)	0.412 (30) → 0.406	0.324 (31) → 0.320	0.403 (27) → 0.400
SaProt (mask)	mutant	0.652 (23) → 0.651	0.465 (30) → 0.460	0.323 (32)	0.461 (30) → 0.458
SaProt (3Di)	masked_wt	0.606 (32)	0.461 (31) → 0.460	0.394 (31) → 0.394	0.454 (31) → 0.453
SaProt (3Di)	unmasked_wt	0.606 (27) → 0.605	0.473 (31) → 0.466	0.415 (31) → 0.406	0.467 (31) → 0.459
SaProt (3Di)	mutant	0.684 (26) → 0.683	0.531 (31) → 0.525	0.451 (31) → 0.445	0.532 (31) → 0.524

3.3. Multi-Scale Pooling Only Appears Beneficial Under the Random Split

We ran the pooling benchmark (mutant arm, GCV-regularized Ridge, fold-internal standardization, attention weights trained per fold) on the 20-dataset design subset for each model under the leakage-free modulo split (Table 4; Figure 3b) and, under the identical pipeline, under the random split. Under leakage-free evaluation no fusion strategy improved on the last layer. For SaProt (structure-masked inputs), all four strategies occupied a narrow band for every readout (Ridge spread across 0.135–0.005; LightGBM 0.014). For ESM-2, spreads were likewise small (Ridge 0.041; LightGBM 0.050), and the best single configuration was the last layer with LightGBM ($\rho = 0.508$), exceeding every multi-scale strategy under every readout. The per-dataset paired differences (Table 3) confirm this: under the modulo split no strategy–readout combination significantly beats the last layer, and with LightGBM several combinations are significantly worse (ESM-2 Mean L20–33 -0.050 ; SaProt Concat 5L -0.012). Under the random split, in contrast, the same comparisons show small but significant advantages for fusion ($+0.009$ to $+0.050$; five of six strategy–readout combinations for SaProt, Concat 5L for ESM-2 under both readouts). The apparent benefit of multi-scale pooling is therefore protocol-dependent: it emerges when positions leak between folds and disappears—occasionally reversing—under leakage-free evaluation. Per-fold-trained attention pooling was never the best strategy under the leakage-free protocol. The full grid of paired

differences across all splits and all four readouts (including contiguous and the RF/MLP readouts) is Supplementary Table S6, and the strategy means under all four splits are Supplementary Table S8.

Table 3. Paired fusion-versus-last-layer differences (mutant arm, combined features, 20 datasets per model). Cells give the mean per-dataset difference $\Delta\rho$ (strategy – last layer) with the bootstrap 90% confidence interval; † marks intervals excluding zero. Under the leakage-free modulo split no strategy significantly beats the last layer, while the random split produces small significant fusion advantages.

Strategy	ESM-2		SaProt	
	Random	Modulo	Random	Modulo
Ridge readout				
Mean L20–33	–0.001 (–0.004, +0.002)	+0.010 (–0.001, +0.023)	+0.003 (–0.001, +0.007)	+0.001 (–0.012, +0.014)
Concat 5L	+0.012 (+0.005, +0.021) †	+0.018 (–0.008, +0.044)	+0.030 (+0.020, +0.039) †	+0.003 (–0.016, +0.021)
Attention	–0.002 (–0.011, +0.007)	–0.024 (–0.053, +0.002)	+0.010 (+0.004, +0.016) †	–0.002 (–0.012, +0.007)
LightGBM readout				
Mean L20–33	–0.002 (–0.005, +0.002)	–0.050 (–0.065, –0.035) †	+0.030 (+0.019, +0.040) †	–0.009 (–0.018, +0.001)
Concat 5L	+0.009 (+0.006, +0.012) †	–0.013 (–0.025, –0.002) †	+0.050 (+0.033, +0.067) †	–0.012 (–0.018, –0.006) †
Attention	–0.002 (–0.010, +0.005)	–0.045 (–0.086, –0.012) †	+0.046 (+0.029, +0.063) †	–0.014 (–0.027, –0.002) †

Table 4. Multi-scale pooling under the unified leakage-free pipeline (mutant arm, modulo split, combined features; 20 datasets per model; mean $\rho \pm$ sd). Spreads across strategies: ESM-2 Ridge 0.041, RF 0.057, MLP 0.045, LightGBM 0.050; SaProt Ridge 0.005, RF 0.010, MLP 0.039, LightGBM 0.014.

ESM-2 650M	Ridge	RF	MLP
Last Layer (baseline)	0.463 $\pm$ 0.174	0.476 $\pm$ 0.173	0.480 $\pm$ 0.178
Mean L20–33	0.473 $\pm$ 0.180	0.420 $\pm$ 0.165	0.436 $\pm$ 0.179
Concat 5L	0.481 $\pm$ 0.183	0.453 $\pm$ 0.181	0.435 $\pm$ 0.170
Attention-weighted	0.439 $\pm$ 0.201	0.419 $\pm$ 0.209	0.440 $\pm$ 0.209
SaProt 650M_AF2	Ridge	RF	MLP
Last Layer (baseline)	0.432 $\pm$ 0.163	0.415 $\pm$ 0.174	0.445 $\pm$ 0.168
Mean L20–33	0.433 $\pm$ 0.166	0.415 $\pm$ 0.166	0.429 $\pm$ 0.166
Concat 5L	0.434 $\pm$ 0.185	0.407 $\pm$ 0.173	0.406 $\pm$ 0.176
Attention-weighted	0.429 $\pm$ 0.164	0.405 $\pm$ 0.166	0.426 $\pm$ 0.164

3.3.1. Readout Choice Interacts With Layer Choice Only for ESM-2

Under the leakage-free protocol, switching the readout on the last layer produced the largest single improvement observed anywhere in the pooling benchmark for ESM-2 (LightGBM over Ridge: paired $\Delta\rho = +0.045$, 90% CI +0.021 to +0.068, Wilcoxon $p = 0.011$), whereas for SaProt the same contrast was not significant (+0.016, 90% CI –0.016 to +0.048, $p = 0.57$; all readout contrasts in Supplementary Table S7). Pooling comparisons that vary the feature construction, scaling, regularization, or split protocol across cells can manufacture much larger spurious contrasts than any of these effects; with the pipeline held fixed, no strategy–readout configuration involving multi-scale features exceeded the best single-layer configuration for either model.

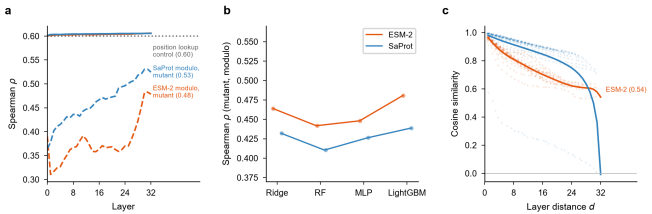


Figure 3 Per-layer probing, multi-scale pooling, and layer distance under a single fixed protocol. (a) 33-layer probing (combined features): under the random split the masked-arm profile is flat for both models ($\sigma \approx 10^{-3}$) and is reproduced by the random-position lookup control (dotted, 0.600); under leakage-free splits (modulo, mutant arm) performance is lower and the best layers are consistently late (layers 30–32, $\Delta\rho \leq 0.009$ versus the final layer). (b) Pooling (mutant arm, modulo split, 20 datasets per model; four strategies as dots, mean connected): ESM-2 spread across strategies 0.041 (Ridge) to 0.057 (RF); SaProt spread 0.005 (Ridge) to 0.039 (MLP); last-layer LightGBM is the best single configuration; per-fold-trained attention pooling is never best. (c) Cosine similarity versus layer distance (unique layer pairs): SaProt decays monotonically and reaches –0.01 only at the largest distances, while ESM-2 remains positive; linear CKA between the most distant SaProt layers remains 0.81 (Figure 2), indicating rotation with preserved linear information rather than loss.

3.4. Logit Lens Confirms Late-Layer Information Accumulation

The lens (fp32, per-dataset validation against the official logits pathway, $n = 63$) produced monotonically improving zero-shot ρ with depth for both models (Figure 4). ESM-2 rose from $\rho = 0.178$ (layer 0) to $\rho = 0.443$ (layer 32), with only small fluctuations in early layers; SaProt with real 3Di rose from $\rho = 0.077$ (layer 0) through approximately zero around layer 20 to $\rho = 0.437$ (layer 32). Both curves attain their maximum at the final layer, agreeing with the leakage-free probing: mutation-relevant information accumulates toward the final layers. Zero-shot absolute levels remain below supervised probing (0.44 versus up to 0.53), so the lens is best read as a consistency check on layer-wise trends rather than a standalone benchmark.

3.5. Agreement With Published ProteinGym Results

The ProteinGym v1.3 score release provides per-assay zero-shot predictions for published models, including ESM-2 650M; 46 of our 63 assays have published scores, and Table 5 places our final-layer results beside the published values recomputed on exactly those 46 assays. Our validated logit-lens final layer gives 0.438 (ESM-2) and 0.440 (SaProt, real-3Di), inside the published 650M-class band on the same assays (ESM-2 650M 0.472, GEMME 0.474, EVE 0.478, DeepSequence 0.457, MSA Transformer 0.447, ESM-1v 0.445, ESM-2 3B 0.448, ESM-2 150M 0.421), and our independent SaProt zero-shot reproduction scores 0.484; the official release contains no per-assay SaProt column, so SaProt is anchored by these two independent scorings of our own. The remaining gap to the official ESM-2 value (0.438 versus 0.472) traces to scoring protocol:

the official scorer masks each mutated position in turn (masked-marginal), whereas the lens scores the full mutant sequence with all mutations in place, a difference that matters because 95.8% of analyzed rows belong to multi-site variants. For supervised anchoring, the same release ships out-of-fold predictions for published supervised methods under the official fold schemes our pipeline implements. Recomputed on our assays: Kermut 0.605, ProteinNPT 0.566, embedding-augmented MSA Transformer 0.572 under the modulo fold (0.45 to 0.56 across methods under contiguous). Our leakage-free probing (0.32 to 0.53) falls inside this published supervised band. The published one-hot baseline without augmentation drops from 0.570 under the random fold to -0.003 under the modulo fold, an external confirmation of the position-memorization mechanism that our lookup control demonstrates (Supplementary Table S10).

Table 5. Published ProteinGym v1.3 results recomputed on the same assays as this work (46 of 63 have published scores). Panel A: zero-shot, mean per-assay Spearman ρ . Panel B: supervised, the release's official out-of-fold predictions under the stated fold scheme. Our rows are matched to the same 46 assays.

A. Zero-shot ($n = 46$)			
Method		ρ	
ESM-2 650M (published)		0.472	
GEMME		0.474	
EVE (single)		0.478	
DeepSequence (single)		0.457	
MSA Transformer (single)		0.447	
ESM-1v (single)		0.445	
ESM-2 3B		0.448	
ESM-2 150M		0.421	
Our lens: ESM-2 final layer		0.438	
Our lens: SaProt (3Di) final layer		0.440	
Our SaProt zero-shot reproduction		0.484	

B. Supervised (official folds, $n = 46$)			
Method	Random	Modulo	Contiguous
Kermut	0.711	0.605	0.561
ProteinNPT	0.697	0.566	0.525
Emb.-aug. MSA Transformer	0.641	0.572	0.536
OHE, not augmented	0.570	-0.003	$+0.027$
Our leakage-free probing (final layers)	—	0.32–0.53	

3.6. Boundary Conditions

The results above localize what the random split does and does not measure. We now examine the two factors that moderate every conclusion: evaluation protocol and input representation.

3.6.1. Leakage-Free Splits: What Is Lost and What Remains

The control and main-table results above already localize the leakage: the random-position lookup table reproduces the random-split level (0.600 versus 0.603–0.606 for full probing) and loses it entirely under leakage-free splits (0.10–0.14), while the random vectors alone carry 0.557 under the random split and approximately zero elsewhere (Tables 1 and 2; Figure 5). Among the leakage-free splits, modulo and position-grouped give similar levels (e.g., SaProt real-3Di mutant arm: 0.531 and 0.532)

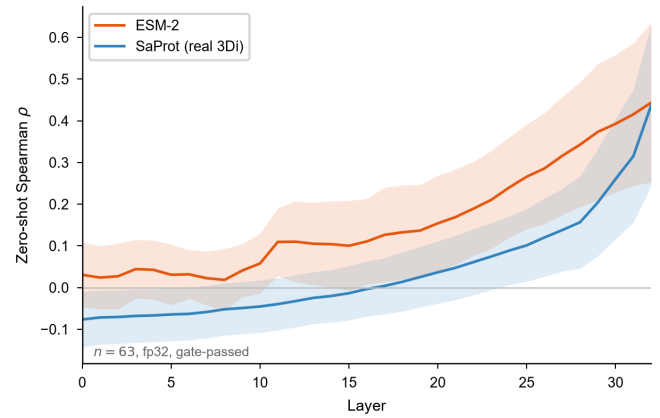


Figure 4 Logit lens: zero-shot mutation scoring by layer ($n = 63$).

Both models improve with depth and attain their maximum at the final layer: ESM-2 (orange) rises from $\rho = 0.03$ (layer 0) to $\rho = 0.443$ (layer 32); SaProt with real 3Di tokens (blue) rises from negative early-layer values to $\rho = 0.437$ (layer 32). Shaded bands indicate ± 1 standard deviation across datasets. All computations in fp32 and validated against the official logits pathway.

while contiguous is uniformly hardest (0.451), consistent with sequence-contiguous segments correlating with local structural context and mutational scope. Random mutation splitting, the de facto standard in DMS benchmarking, therefore inflates absolute performance estimates by roughly 0.15–0.20 ρ in these arms, masks both layer differences and structure-token contributions, and can make multi-scale fusion appear beneficial (Section 3.3). ProteinGym itself defines the modulo and contiguous leakage-free splits used here, so reporting one of them alongside the standard random split requires no new infrastructure—analogueous to the time-series community's practice of reporting both random and temporal cross-validation.

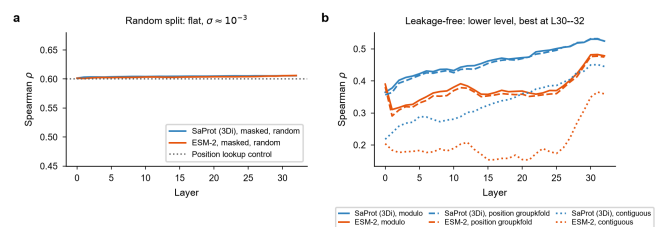


Figure 5 Random versus leakage-free split probing (SaProt, 63 datasets). (a) Random mutation split: flat per-layer profile ($\sigma \approx 10^{-3}$), reproduced by the random-position lookup control ($\rho = 0.600$). (b) Leakage-free splits (modulo, position-grouped, contiguous): performance drops to 0.27–0.53 depending on arm and model (Table 2; minimum 0.274, SaProt structure-masked masked_wt under contiguous), the invariance score rises to $4\text{--}7 \times 10^{-2}$, and the best layers shift consistently to layers 30–32.

Together, these boundary conditions refine the central findings: apparent layer interchangeability under random splitting is dominated by position memorization, leakage-free evaluation reveals a consistent depth trend whose final layers are

sufficient, the apparent benefit of multi-scale fusion is protocol-dependent, and the benefit of structure tokens emerges mainly under leakage-free evaluation.

3.7. Structure Tokens Help Mainly Under Leakage-Free Evaluation

Because SaProt can be run with real 3Di tokens or with structure-masked tokens, the design allows a paired test, dataset by dataset and layer by layer, of what structure tokens actually contribute (Table 6). Under the random split the two input modes were nearly identical (paired differences at layer 32: +0.003 masked_wt, +0.002 unmasked_wt, +0.032 mutant). Under every leakage-free split the real-3Di inputs were consistently superior, with mean paired advantages of +0.059 to +0.122 across arms and splits, all with Wilcoxon $p \leq 10^{-4}$. Structure tokens therefore carry genuine mutation-relevant information: the contribution is already positive under random splitting (+0.003 to +0.032, significant for the mutant arm) and becomes large when position memorization is removed from the evaluation. This also explains why structure-aware inputs add little in random-split comparisons: the position channel, available to both input modes, saturates most of the task.

Table 6. Real-3Di versus structure-masked SaProt inputs (paired per-dataset differences at layer 32, combined features; Wilcoxon signed-rank p). The benefit of structure tokens is small and mostly non-significant under the random split and large and highly significant under all leakage-free splits.

Arm	Random		Modulo	Leakage-free	
	$\Delta\rho$	p		Contiguous	Pos.-grouped
masked_wt	+0.003	2.2×10^{-2}	+0.073 (3.5×10^{-7})	+0.120 (8.7×10^{-6})	+0.071 (8.1×10^{-6})
unmasked_wt	+0.002	9.1×10^{-1}	+0.060 (4.6×10^{-6})	+0.086 (2.1×10^{-6})	+0.059 (1.2×10^{-5})
mutant	+0.032	2.5×10^{-6}	+0.064 (6.9×10^{-5})	+0.122 (6.5×10^{-7})	+0.067 (4.6×10^{-5})

3.8. Sensitivity Analysis: Single-Substitution Variants Only

Because 95.8% of analyzed rows belong to multi-site variants whose rows can share fitness labels across the official position-based folds (Section 2.1), we re-ran the probing experiment for all three model configurations restricted to single-substitution variants: one row per variant, and no variant-level label sharing across folds remains. The depth trend and the late-band optimum are unchanged—the best layer is L30 or L31 in every leakage-free column (Supplementary Table S11). Final-layer sufficiency is preserved for SaProt, where the best-versus-final difference stays within $\Delta\rho \leq 0.010$ (structure-masked ≤ 0.006 , all bootstrap 90% confidence intervals overlapping zero; real-3Di ≤ 0.010). For ESM-2 the gap widens modestly: best-minus-final reaches +0.016 (modulo; 90% CI +0.006 to +0.027) and +0.019 (position-grouped; CI +0.004 to +0.036), so on single-substitution data the ESM-2 final layer concedes up to about 0.02 ρ to layers 30–31, slightly beyond the 0.01 full-benchmark margin. The structure-token benefit replicates, and strengthens, on single-site data: real-3Di over structure-masked at the final

layer gives +0.028 under random ($p = 4.3 \times 10^{-5}$) and +0.057 / +0.132 / +0.058 under modulo / contiguous / position-grouped (all $p \leq 10^{-3}$). Finally, an Envision-style supervised baseline (one-hot and substitution features with MSA PSSM columns where available, gradient-boosted readout; 50 of 63 assays with usable MSAs) shows the same protocol signature externally: $\rho = 0.634$ under random versus 0.410 modulo and 0.302 contiguous (Supplementary Table S10).

4. Discussion

The central result of this study has two halves. Under the standard random-split protocol, the 33 transformer layers of SaProt and ESM-2 show near-flat mutation effect prediction performance, and the position-lookup control shows that this flatness is largely position memorization rather than distributed mutation information. Under leakage-free evaluation the picture inverts: mutation-specific performance rises with depth (ρ up to 0.53), the best layers are consistently the last few, the final layer is within 0.01 ρ of the best layer in every full-benchmark comparison (within 0.02 on a single-substitution-only subset for ESM-2), and structure tokens deliver a large measurable benefit (+0.06 to +0.12). Multi-scale pooling provides no gain under leakage-free evaluation for either model; its apparent benefit appears only under the leaking protocol. The practical conclusion is not that multi-scale embeddings are never useful, but that for mutation effect prediction with 650M PLMs the final layer is sufficient, and that the evaluation protocol determines whether a benchmark can see the difference.

4.1. Why Does Depth Improve Mutation Effect Prediction?

The depth trend under leakage-free evaluation is consistent with the MLM pre-training objective. All layers receive gradients through the same cross-entropy loss computed at the final LM head, so each successive layer is optimized to move representations closer to the quantities the LM head must predict—conditional amino-acid distributions—which are what a mutation-effect probe ultimately reads out. Layers therefore accumulate, rather than specialize, mutation-relevant signal, and the last layers carry the most of it. Consistent with accumulation rather than specialization, the residual layer variation under leakage-free evaluation has a consistent direction (later is better) instead of task-dependent optima at different depths.

The cosine similarity between layers decays monotonically with distance and, for SaProt, collapses to approximately zero only between the most distant layers, while the linear CKA between the same layers remains 0.78. This combination, a near-rotation that preserves linearly decodable information, is consistent with the effect of residual connections : because residual streams accumulate representations from all preceding layers, late layers retain a linearly accessible trace of early-layer content even as their raw coordinates rotate.

This interpretation is consistent with the logit lens hypothesis , , which posits that intermediate layers in transformer language models progressively refine token predictions toward the final output distribution, rather than constructing qualitatively different representations. The lens curves support

this directly for both models: zero-shot scoring improves with depth and peaks at the final layer. The near-orthogonality of the most distant SaProt layers ($\cos \approx -0.01$) alongside their high linear CKA (0.78) suggests that mutation-relevant signal remains accessible while representations move through successive coordinate frames.

Notably, this pattern contrasts with findings from NLP transformers, where probing classifiers consistently recover different linguistic features from different layer depths. In BERT, for example, surface syntactic features peak in early layers while semantic features dominate in middle-to-late layers. The single-peaked, monotonic profile of the PLM curves may reflect the fundamentally different nature of the learning signal in protein versus natural language: whereas language contains hierarchical structure at multiple timescales (morphology, syntax, semantics, discourse), the MLM task on protein sequences operates on a single level of abstraction (amino acid identity), potentially eliminating the pressure for mid-network specialization.

4.1.1. What Separates SaProt from Standard ESM-2?

Under the unified protocol the two models differ mainly in two measured ways: how much their final representations retain early-layer geometry (ESM-2 distant-layer cosine 0.54 with CKA 0.41; SaProt cosine -0.008 with CKA 0.78), and the benefit conferred by structure tokens. The directly measured difference is the conditional structure-token benefit: real 3Di inputs improve leakage-free probing by $+0.06$ to $+0.12$ while leaving random-split performance nearly unchanged. Structure information is genuinely present and genuinely useful, but leakage-free evaluation is required to measure it.

This framing carries a testable prediction: fine-tuning standard ESM-2 with structural objectives (e.g., contact prediction or fold classification) should reproduce a similar leakage-free advantage over the sequence-only baseline. This remains a mechanistic hypothesis rather than a direct result of the present experiments.

4.2. Reconciling with Prior Layer-Selection Studies

Our findings do not contradict the observation that different layers may be optimal for different task types. Kumar and Jha demonstrated that mid-to-late ESM-2 layers (20–33) outperform the final layer for kinase function prediction, achieving 32% higher clustering quality and superior classification accuracy. This is compatible with our narrower conclusion: different downstream tasks may induce different optimal readout directions, while mutation effect prediction under random-split probing shows little layer dependence. The key distinction is between task-level specialization and within-task layer complementarity under a specified readout and evaluation protocol.

Similarly, the 12% Spearman improvement reported by Alsamkary, Elshaffei et al. for attention-based fusion architectures was observed in the context of multi-chain protein–protein interaction prediction, a task that differs substantially from per-residue mutation effect scoring. The benefit of sophisticated pooling may therefore be task-dependent rather than inherent to multi-scale representation fusion. For mutation effect prediction,

our results suggest that such improvements should be interpreted in light of model type, readout choice, and split protocol: under leakage-free evaluation no fusion strategy improved on the final layer for either model, and the small fusion advantages that appear under random splitting track the position-leakage regime rather than additional mutation information (Section 3.3).

4.3. Practical Implications

For researchers using protein language model embeddings for mutation effect prediction, our results support two practical recommendations that depend on the model variant, readout model, and evaluation protocol:

- Structure-aware models (SaProt, ESM-2 fine-tuned with structural objectives):** The final transformer layer is a strong default. In our benchmarks, multi-scale fusion provided no practical benefit over single-layer embeddings for either model under leakage-free evaluation, and 1280 dimensions sufficed for both linear and nonlinear readouts. This reduces storage and computational costs by a factor of 5–10 compared to the common practice of concatenating 6–8 layers (up to 10,240 dimensions).
- Standard PLMs (ESM-2, ProtBERT, etc.):** the final layer with a nonlinear readout is the strongest configuration we measured: last-layer LightGBM reached $\rho = 0.508$ ($+0.045$ over the same layer with Ridge, $p = 0.011$), the best pooling configuration overall. Layer concatenation offered no reliable gain under leakage-free evaluation ($+0.018$ with Ridge, not significant, and negative with LightGBM) at $4.9\times$ the feature dimension. For SaProt the readout contrast was not significant ($+0.016$, $p = 0.57$).

Table 7 quantifies the practical trade-offs. All pooling strategies require one forward pass per variant with all 33 layers collected, so embedding-extraction cost is identical across strategies and dominates the end-to-end time (~ 3 minutes per dataset on an RTX 5090 for every strategy). What Concat 5L inflates is the readout: its covariance matrix grows from $1,322^2$ to $6,442^2$ entries (a $24\times$ penalty on the Ridge solve) and its per-mutation storage grows $4.8\times$ —in exchange for no benefit under leakage-free evaluation on either model, and a configuration (ESM-2 last-layer LightGBM) that exceeds it on a single layer. Storing Concat 5L embeddings for a typical DMS benchmark ($\sim 10,000$ variants) requires 258 MB versus 53 MB for a single layer, a gap that widens linearly with dataset size and becomes substantial for genome-scale studies.

In both cases, computational resources may be better directed toward feature types that are plausibly complementary to PLM embeddings:

- Structural features (e.g., AlphaFold-predicted pLDDT, PAE, or Foldseek 3Di states), which complement sequence-based embeddings;
- Sequence-level representations that capture global protein properties beyond per-residue features;
- Multiple sequence alignment (MSA)-based approaches, that leverage evolutionary information beyond what PLMs encode.

Table 7. Computational cost comparison across pooling strategies. “Per-mutation” storage assumes float32 precision. Time estimates are per-dataset averages on an NVIDIA RTX 5090, including embedding extraction and readout training across 20 ProteinGym DMS subsets. **Best achievable ρ from Table 4 (mutant arm, modulo split).**

	Last Layer	Mean L20–33	Attention	Concat 5L
Feature dimension	1,322	1,322	1,322	6,442
Storage per mutation (KB)	5.2	5.2	5.2	25.2
Ridge design matrix (5,000 vars)	6.6×10^6	same	same	3.2×10^7
Ridge covariance matrix entries	1.7×10^6	same	same	4.1×10^7
Avg. time per dataset (min)	~3	~3	~3	~3
Best achievable ρ (ESM-2 / SaProt):				
With Ridge	0.463 / 0.432	0.473 / 0.433	0.439 / 0.429	0.481 / 0.434
With LightGBM	0.508 / 0.448	0.457 / 0.439	0.463 / 0.433	0.495 / 0.436

As discussed under Boundary Conditions (Section 3.5), the position-lookup control shows that standard evaluation protocols can conflate mutation-effect learning with position-level memorization. We recommend that future DMS benchmarks adopt position-aware evaluation as a standard complement to random splitting.

4.4. Limitations

Several limitations should be noted. First, we tested the 650M-parameter ESM-2 variant and its structure-aware derivative SaProt. Smaller (35M, 150M) and larger (3B, 15B) variants may show different degrees of layer specialization, and this remains to be tested. Second, our experiments are restricted to per-residue mutation effect prediction; for other tasks such as kinase function prediction, subcellular localization, or protein–protein interaction scoring, layer-wise differences and the benefits of multi-scale fusion may be more pronounced. Third, our pooling benchmark used a 20-dataset validation subset, which is smaller than the 63-dataset probing analysis. Fourth, the SaProt pooling runs use structure-masked inputs, whereas the probing and logit lens runs include real 3Di inputs, and one dataset (influenza nucleoprotein) lacks an AFDB structure. Fifth, the official ProteinGym modulo and contiguous splits are position-based, so the rows of a multi-site variant—95.8% of all analyzed rows—can share a fitness label across folds; these splits remove position leakage but not variant-level label sharing, and our single-substitution sensitivity analysis (Section 3.8) addresses exactly this caveat. Sixth, our feature ablation (embedding-only vs. auxiliary-only) indicates that learned embeddings dominate predictive performance in this setting, but it does not rule out the possibility that nonlinear combinations of auxiliary features could provide stronger baselines. Seventh, the maximum sample size of 10,000 variants per dataset (subsampled to 5,000 rows for regression; 11 of 63 datasets were variant-capped and 26 reach the row ceiling) may limit sensitivity to subtle layer-wise effects on the largest DMS screens (>200,000 mutations), though the consistency of layer profiles across dataset-size categories argues against this concern. Eighth, our linear probing approach can only detect information that is linearly separable; nonlinear interactions between layers could theoretically exist but would require a different experimental design to identify. Ninth, small-sample CKA is upward-biased at our dimensionality (50-position ESM-2 distant-layer CKA 0.55 versus 0.41 at 200 positions), and the debiased unbiased-HSIC estimator is unstable when $n < d$ (point estimates can exceed 1); we therefore

report the larger-sample standard estimator. Tenth, position-level data splitting (Section 3.5) shows that random mutation splits permit position-level leakage, which the lookup control quantifies and which motivates our reporting of leakage-free splits throughout. Finally, our equivalence margin ($\Delta\rho = 0.01$) was fixed before the equivalence analysis as the smallest difference that could plausibly change a downstream layer-choice decision; it is a reporting convention, not a task-derived requirement, and it is applied identically everywhere. The best layer in each comparison is selected on the same per-dataset results, which biases the procedure towards detecting a difference and therefore makes the equivalence verdicts conservative.

4.5. Conclusion

For mutation effect prediction with 650M protein language models, depth matters: under leakage-free evaluation, predictive information rises monotonically through the network and the best layers are the last ones. But the final layer is already sufficient—it is within 0.01 ρ of the best layer in every full-benchmark comparison we ran (and within 0.02 on a single-substitution-only subset for ESM-2), and no multi-scale fusion strategy improved on it; the small fusion advantages that do appear are artifacts of the position-leaking random split, the same regime in which a position lookup table with no sequence information reproduces the apparent layer interchangeability of the literature. Structure tokens deliver a genuine, measurable benefit (+0.06 to +0.12) that only leakage-free evaluation can see. Together these results replace the multi-scale-fusion heuristic for this task with a simpler rule—extract the final layer, invest in the readout, and evaluate under a position-aware split—and they argue for treating claims of layer-specific representations in PLMs with the same scrutiny that has become standard in the NLP community.

Acknowledgments

We thank the International College of Pharmaceutical Innovation, SUDA-RCSI for institutional support.

Funding

This work was supported by the Jiangsu Tepin Foundation (Z241922) and the Health Research Board (ILP-PHR-2024-014).

Competing Interests

The authors declare no competing interests.

Data Availability

Code and results are available at <https://github.com/schwarzteeguo-creator/esm2-layer-invariance>. An archived snapshot of the code and per-dataset results for this revision is deposited at Zenodo (concept DOI 10.5281/zenodo.22844290; archived version v3.0.0, DOI 10.5281/zenodo.23264208). Raw ProteinGym data: <https://github.com/OATML-Markslab/ProteinGym>.

References

- Rives, A. *et al.* (2021) Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences*, **118**(15), e2016239118.
- Meier, J. *et al.* (2021) Language models enable zero-shot prediction of the effects of mutations on protein function. *Advances in Neural Information Processing Systems*, **34**, 29287–29303.
- Lin, Z. *et al.* (2023) Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, **379**(6637), 1123–1130.
- Frazer, J. *et al.* (2021) Disease variant prediction with deep generative models of evolutionary data. *Nature*, **599**(7883), 91–95.
- Hie, B. *et al.* (2021) Learning the language of viral evolution and escape. *Science*, **371**(6526), 284–288.
- Elnaggar, A. *et al.* (2022) ProtTrans: Toward understanding the language of life through self-supervised learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, **44**(10), 7112–7127.
- Brandes, N. *et al.* (2022) ProteinBERT: A universal deep-learning model of protein sequence and function. *Bioinformatics*, **38**(8), 2102–2110.
- Rogers, A., Kovaleva, O. and Rumshisky, A. (2020) A primer in BERTology: What we know about how BERT works. *Transactions of the Association for Computational Linguistics*, **8**, 842–866.
- Tenney, I., Das, D. and Pavlick, E. (2019) BERT rediscovers the classical NLP pipeline. *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 4593–4601.
- nostalgebraist (2020) Interpreting GPT: the logit lens. <https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/>.
- Kumar, A. and Jha, I.P. (2025) Layer probing improves kinase functional prediction with protein language models. *arXiv*, 2512.00376.
- Detlefsen, N.S. *et al.* (2022) Learning meaningful representations of protein sequences. *Nature Communications*, **13**(1), 1914.
- Alsamkary, H., Elshaffei, M. *et al.* (2025) Beyond simple concatenation: Fairly assessing PLM architectures for multi-chain protein–protein interactions prediction. *arXiv*, 2505.20036.
- Rao, R. *et al.* (2019) Evaluating protein transfer learning with TAPE. *Advances in Neural Information Processing Systems*, **32**.
- Notin, P. *et al.* (2024) ProteinGym: Large-scale benchmarks for protein fitness prediction and design. *Advances in Neural Information Processing Systems*, **36**.
- Su, J. *et al.* (2024) SaProt: Protein language modeling with structure-aware vocabulary. *International Conference on Learning Representations (ICLR)*.
- van Kempen, M. *et al.* (2024) Fast and accurate protein structure search with Foldseek. *Nature Biotechnology*, **42**(2), 243–246.
- Roberts, D.R. *et al.* (2017) Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, **40**(8), 913–929.
- He, K. *et al.* (2016) Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778.
- Jumper, J. *et al.* (2021) Highly accurate protein structure prediction with AlphaFold. *Nature*, **596**(7873), 583–589.
- Riesselman, A.J., Ingraham, J.B. and Marks, D.S. (2018) Deep generative models of genetic variation capture the effects of mutations. *Nature Methods*, **15**(10), 816–822.
- Hopf, T.A. *et al.* (2017) Mutation effects predicted from sequence co-variation. *Nature Biotechnology*, **35**(2), 128–135.
- Golub, G.H., Heath, M. and Wahba, G. (1979) Generalized cross-validation as a method for choosing a good ridge parameter. *Technometrics*, **21**(2), 215–223.
- Kornblith, S., Norouzi, M., Lee, H. and Hinton, G. (2019) Similarity of neural network representations revisited. *Proceedings of the 36th International Conference on Machine Learning*, 3519–3529.
- Belrose, N. *et al.* (2023) Eliciting latent predictions from transformers with the tuned lens. *arXiv*, 2303.08112.
- Székely, G.J., Wattenberg, M. and Sundararajan, M. (2021) Feature space or sample space? A debiased kernel regression estimator. *Advances in Neural Information Processing Systems*, **34**, 7154–7164.
- Gray, V.E., Hause, R.J., Luebeck, J., Shendure, J. and Fowler, D.M. (2018) Quantitative missense variant effect prediction using large-scale mutagenesis data. *Cell Systems*, **7**(1), 136–144.
- Ko, Y.S., Parkinson, J. and Wang, W. (2026) Scalable embedding fusion with protein language models: insights from benchmarking text-integrated representations. *Briefings in Bioinformatics*, **27**(1), bbag014.
- Si, Y. and Yan, C. (2024) Protein language model-embedded geometric graphs power inter-protein contact prediction. *eLife*, **12**, RP92184.